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DA - 02
Design & Analysis Algorithm
Karatsuba

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Que → Develop an algorithm using divide and conquer strategy to multiply two integers 88409 and 83329. Generate the recurrence relation from the algorithm and compute the time complexity. Also compute how much time it will take to multiply the given two integers.

Solve- $x = 88409$ and $y = 83329$
make same length,

$x = 088409$, $y = 083329$ $(n = 6)$
and even.

finding $(n/2)$, $n/2 = 3$, $10^{n/2} = 10^3$

$x = a \times 10^3 + b$, $y = c \times 10^3 + d$
$a = 88$, $b = 409$, $c = 83$, $d = 329$

Applying Karatsuba formula :

$$xy = ac \times 10^{2(n/2)} + ((a+b)(c+d) - ac - bd) * 10^{n/2} + bd$$

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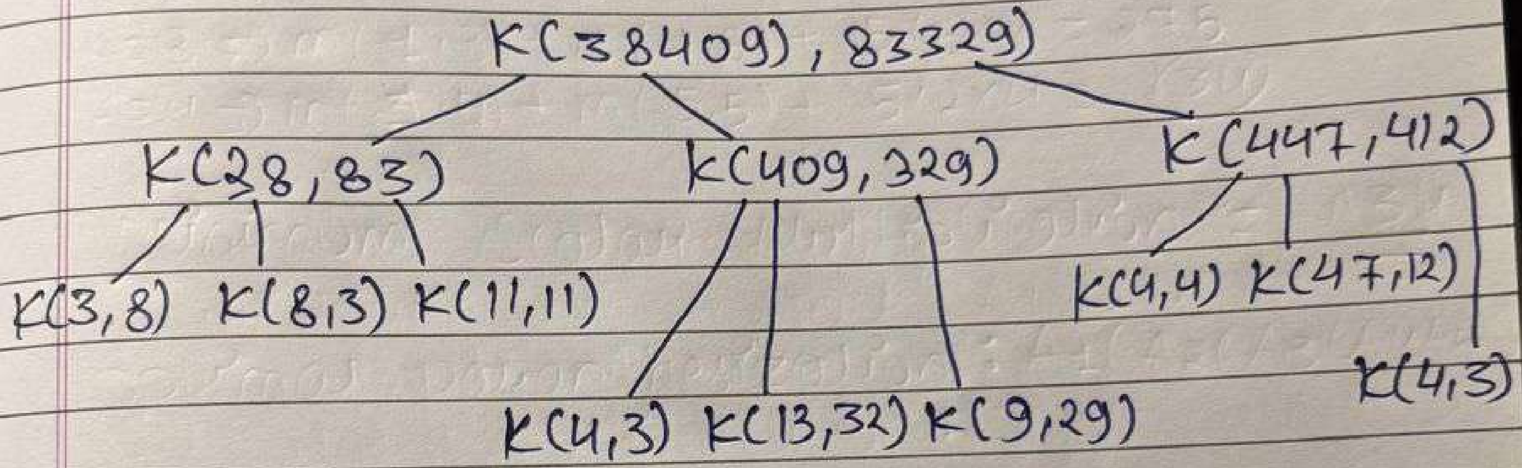
$= 38\times 83\times 10^{n} + (184164 - 3154 - 134561)\times 10^{n/2}$
$+ 134561$

$= 3154\times 10^{6} + 46449\times 10^{3} + 134561$
$= 3154000000 + 46449000 + 134561$
$(xy) = 3200583561$

* Recurrence Tree :-



2) Recurrence Relation :- $T(n) = 3T(\frac{n}{2}) + O(n)$

2) Time Complexity :- Using Master's Theorem :

$$T(n) = O(n^{\log_2 3}) \approx O(n^{1.585})$$

2) Since Both integers contains 5 digits the time taken to multiply them Using Divide And Conquer

$$O(5^{1.585}) \approx O(13)$$ Ans